Samar Pratap

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Education

VIT Bhopal University — **Sehore, MP**
Bachelor of Technology in Computer Science and Engineering — **Expected 2027**
CGPA: 8.32

PM Shri Kendriya Vidyalaya No1 HBK Salawala — **Dehradun, Uttarakhand**
Higher Secondary School (PMC+CS) — **Graduated July 2022**
Percentage: 88.2%

Technical Skills

Languages: Python, C/C++, Java, SQL, JavaScript, HTML/CSS, Bash

Frameworks & Libraries: Pandas, NumPy, NLTK, BeautifulSoup, Streamlit, NetworkX, Pyvis, Tkinter, Node.js, Pygame-ce

Databases: MySQL

Developer Tools: Git, VS Code, Linux, Vim, Obsidian

Project Experience

Social Media Sentiment Analysis Dashboard | *Python, NLP, Pandas, NLTK, BeautifulSoup, PRAW, Streamlit*
Engineered an end-to-end NLP data pipeline that scrapes large-scale Reddit discussions, applies VADER sentiment analysis, and visualizes insights through an interactive Streamlit dashboard.

- Developed an automated web-scraping system using BeautifulSoup and Reddit API to collect and preprocess real-world discussion data.
- Implemented NLP-based sentiment analysis to quantify user opinions and extract meaningful trends.
- Designed a responsive dashboard with Streamlit to present interactive visual analytics.

Arcane Signatures: D&D Class Spell Analysis | *Python, Pandas, NetworkX, Pyvis*
A data analysis and visualization tool to identify unique magical signatures of D&D classes.

- Engineered a network graph using NetworkX and Pandas to model complex spell relationships.
- Developed an interactive visualization with Pyvis using ForceAtlas2 for clarity and analysis.

Book Vault: Library Management System | *HTML/CSS, JavaScript, Node.js, JSON*
A web-based library management system developed as a team project.

- Built backend logic using Node.js and JSON for data persistence.
- Integrated an AI chatbot and rule-based book recommendation system.

Spaceship Shooter Game | *Python, Pygame-ce*
A 2D arcade game featuring collision mechanics and scoring systems.

- Implemented pixel-perfect collision detection.
- Designed responsive restart and end-game interfaces.

Python MP3 Player | *Python, Tkinter, Pygame*
A desktop MP3 player application with playlist management.

- Developed playback controls and dynamic playlist features.
- Implemented an interactive volume control slider.